

MATH 3150 — Quiz 1 Study Guide

Methods, templates, worked examples, and practice problems

(Weeks 1–4: $\mathbb{N}, \mathbb{Q}, \mathbb{R}$, induction, sup/inf, completeness, absolute value, limits of sequences, monotone sequences)

Quiz 1: Tuesday, September 29

Contents

1	How to use this guide	3
2	Logic and proof techniques	4
2.1	The basic proof moves	4
2.2	Negating quantifiers	4
2.3	Double inclusion (HW1 P1)	5
2.4	Induction (HW1 P2, P3)	5
2.5	Contradiction and irrationality (HW1 P4, Week 2)	8
3	Supremum, infimum, completeness	9
3.1	Definitions (state these exactly)	9
3.2	Archimedean property and density	10
3.3	Computing sup and inf of explicit sets	10
3.4	Standard sup/inf proofs: templates	12
3.5	Monotonicity of sup	13
4	Absolute value	13
5	Limits of sequences: the ε–N definition	14
5.1	Limits equal to $\pm\infty$	17
6	Limit theorems (Week 3–4)	18
6.1	The quotient rule and the “bounded away from zero” lemma	19
6.2	More limit tools	19
6.3	Binomial-theorem estimates (Week 4 style)	19
6.4	Basic limits to memorize	21
7	Monotone sequences and the Monotone Convergence Theorem	22
8	Preview: subsequences and Bolzano–Weierstrass (Week 5)	25
9	Common mistakes that cost points	25
10	Practice problems	26

1 How to use this guide

Each section follows the same pattern: **definitions you must be able to state word for word**, then **the method**, then **worked examples**. Section 10 has practice problems, and their solutions are in Section 11. Try each problem on paper before you look at the solution.

Quiz scope (from the syllabus).

- HW1: set identities, induction, rational/irrational arguments.
- HW2: density of $\mathbb{Q}$ and of the irrationals, sup/inf, the completeness axiom, $\sup(A + B)$.
- HW3: triangle and reverse triangle inequality, ε - N limits, $x_n \rightarrow \infty$, monotone sequences.
- Week 4 lectures: limit laws, basic limits, limits equal to $\pm\infty$, the Monotone Convergence Theorem.

The single most important skill: match the *shape* of the claim to the proof technique.

Claim looks like...	Reach for...
$S = T$ (sets)	Double inclusion, element chasing
“for all $n \geq n_0$ ”	Induction
“ x is irrational”, “no such ...”	Contradiction
“is it always true?” (suspect false)	One counterexample
$M = \sup S$	(i) M is an upper bound, (ii) $\forall \varepsilon > 0 \exists s \in S, s > M - \varepsilon$
$\sup A \leq \text{something}$	Show that “something” is an upper bound of A
$\lim x_n = L$	ε - N : scratch work, then a forward write-up
$\lim x_n = \infty$	$\forall M \exists N: x_n > M$ for $n \geq N$
x_n converges (limit unknown)	Monotone Convergence Theorem
$ a \leq c$	Show $-c \leq a \leq c$

Definitions cheat sheet (memorize word for word)

Upper bound	M with $s \leq M$ for all $s \in S$
$\sup S$	upper bound M such that no $M' < M$ is an upper bound $\iff$ upper bound and $\forall \varepsilon > 0 \exists s \in S : s > M - \varepsilon$
$\inf S$	lower bound m such that no $m' > m$ is a lower bound $\iff$ lower bound and $\forall \varepsilon > 0 \exists s \in S : s < m + \varepsilon$
Completeness	nonempty + bounded above $\Rightarrow$ sup exists in $\mathbb{R}$
Archimedean	$\forall x \in \mathbb{R} \exists n \in \mathbb{N} : n > x$; equivalently $\forall \varepsilon > 0 \exists n : \frac{1}{n} < \varepsilon$
Density	$x < y \Rightarrow \exists r \in \mathbb{Q}$ and $\exists z \notin \mathbb{Q}$ with $x < r, z < y$
$ x \leq a$	$\iff -a \leq x \leq a$ (for $a \geq 0$)
$x_n \rightarrow L$	$\forall \varepsilon > 0 \exists N \forall n \geq N : x_n - L < \varepsilon$
$x_n \not\rightarrow L$	$\exists \varepsilon > 0 \forall N \exists n \geq N : x_n - L \geq \varepsilon$
$x_n \rightarrow \infty$	$\forall M \exists N \forall n \geq N : x_n > M$
Bounded seq.	$\exists M > 0 \forall n : x_n \leq M$
Increasing	$x_n \leq x_{n+1}$ for all n (decreasing: $x_n \geq x_{n+1}$)
MCT	bounded + monotone $\Rightarrow$ convergent (to sup or inf of the terms)
Cauchy	$\forall \varepsilon > 0 \exists N \forall n, m \geq N : x_n - x_m < \varepsilon$

2 Logic and proof techniques

2.1 The basic proof moves

- **Direct:** assume the hypotheses and chain implications to the conclusion.
- **Contrapositive:** $P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$. Example: “ n^2 even $\Rightarrow n$ even” is easiest as “ n odd $\Rightarrow n^2$ odd”.
- **Contradiction:** assume $\neg(\text{claim})$ and derive something false.
- **Proving “ $a \leq b$ ” via ε :** if $a < b + \varepsilon$ for every $\varepsilon > 0$, then $a \leq b$. (Otherwise take $\varepsilon = a - b > 0$.) Likewise, if $|a - b| < \varepsilon$ for every $\varepsilon > 0$, then $a = b$. **This is how uniqueness of limits and many sup facts are proved.**
- **“For every” statements:** begin with “Let ... be arbitrary.” **“There exists” statements:** construct or exhibit the object.

2.2 Negating quantifiers

To negate a statement, swap $\forall \leftrightarrow \exists$ and negate the inner condition.

$$\neg(\forall \varepsilon > 0 \exists N \forall n \geq N : |x_n - L| < \varepsilon) \iff \exists \varepsilon > 0 \forall N \exists n \geq N : |x_n - L| \geq \varepsilon.$$

So “ $x_n \not\rightarrow L$ ” means: *some* ε is a bad tolerance, and the sequence keeps returning to distance $\geq \varepsilon$ from L no matter how far out you go. You use this to prove divergence (Example 5.8).

2.3 Double inclusion (HW1 P1)

Method 2.1. To prove $S = T$, prove $S \subseteq T$ and $T \subseteq S$. For $S \subseteq T$: “Let $x \in S$ be arbitrary. ... Therefore $x \in T$.” Split into cases wherever the definition involves “or” (a union) or “not (and)” (a complement of an intersection).

Example 2.2 (De Morgan-type identity). $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$.

Quiz-length version. ($\subseteq$) Let $x \in A \setminus (B \cup C)$. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$.

($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$. Then $x \in A$, $x \notin B$, and $x \notin C$, so $x \notin B \cup C$. Hence $x \in A \setminus (B \cup C)$. $\square$

Full proof, step by step. Step 1 (unpack the definitions you will use).

$$x \in A \setminus D \iff (x \in A \text{ and } x \notin D), \quad x \in D \cup E \iff (x \in D \text{ or } x \in E).$$

Negating the second one gives $x \notin D \cup E \iff (x \notin D \text{ and } x \notin E)$. This negation of an “or” is the whole content of the identity.

Step 2 ($\subseteq$: start with an arbitrary element). Let $x \in A \setminus (B \cup C)$ be arbitrary. By the definition of set difference, $x \in A$ and $x \notin B \cup C$.

Step 3 (negate the union). By Step 1, $x \notin B \cup C$ means $x \notin B$ and $x \notin C$.

Step 4 (repackage). From $x \in A$ and $x \notin B$ we get $x \in A \setminus B$. From $x \in A$ and $x \notin C$ we get $x \in A \setminus C$. So x is in both, i.e. $x \in (A \setminus B) \cap (A \setminus C)$. Since x was arbitrary, $A \setminus (B \cup C) \subseteq (A \setminus B) \cap (A \setminus C)$.

Step 5 ($\supseteq$: arbitrary element of the right side). Let $x \in (A \setminus B) \cap (A \setminus C)$ be arbitrary. By the definition of intersection, $x \in A \setminus B$ and $x \in A \setminus C$, so $x \in A$, $x \notin B$, and $x \notin C$.

Step 6 (rebuild the union). Since $x \notin B$ and $x \notin C$, Step 1 gives $x \notin B \cup C$. Together with $x \in A$, this says $x \in A \setminus (B \cup C)$. So $(A \setminus B) \cap (A \setminus C) \subseteq A \setminus (B \cup C)$.

Step 7 (conclude). Both inclusions hold, so the sets are equal. $\square$

Why no cases here? There is no “or” among the *assumptions*. The only “or” is inside $x \notin B \cup C$, and negating it turns it into an “and”. In Practice Problem 4 the “or” is an assumption, so you must split into cases.

2.4 Induction (HW1 P2, P3)

Method 2.3. 1. **Base case:** check $P(n_0)$ directly.

2. **Write out** $P(n+1)$ by replacing every n with $n+1$. This is your target, so write it down first.
3. **Inductive step:** assume $P(n)$. Find the piece of $P(n+1)$ that is exactly $P(n)$ and substitute the hypothesis.
4. Finish with algebra.

If the *starting* index of a sum also moves with n , add and subtract the boundary terms that don’t match (HW1 P3 trick).

Example 2.4 (Bernoulli’s inequality, used in Week 4). For $x \geq -1$ and $n \in \mathbb{N}$: $(1+x)^n \geq 1+nx$.

Proof. Base case $n = 1$: $1+x \geq 1+x$. $\checkmark$

Inductive step: assume $(1+x)^n \geq 1+nx$. Since $1+x \geq 0$, we may multiply both sides by $1+x$ without flipping the inequality:

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

because $nx^2 \geq 0$. □

Full proof, step by step. Fix $x \geq -1$. Let $P(n)$ be the statement $(1+x)^n \geq 1+nx$. We prove $P(n)$ for all $n \in \mathbb{N}$ by induction.

Step 1 (base case $n = 1$). LHS = $(1+x)^1 = 1+x$ and RHS = $1 + 1 \cdot x = 1+x$. Since $1+x \geq 1+x$, $P(1)$ holds. ✓

Step 2 (write the target $P(n+1)$).

$$(1+x)^{n+1} \geq 1 + (n+1)x.$$

Step 3 (peel off one factor). Assume $P(n)$. Then $(1+x)^{n+1} = (1+x)^n \cdot (1+x)$.

Step 4 (use the hypothesis $x \geq -1$). Since $x \geq -1$, we have $1+x \geq 0$. Multiplying both sides of $P(n)$ by the nonnegative number $1+x$ keeps the direction of the inequality:

$$(1+x)^n(1+x) \geq (1+nx)(1+x).$$

Step 5 (expand).

$$(1+nx)(1+x) = 1 + x + nx + nx^2 = 1 + (n+1)x + nx^2.$$

Step 6 (drop the extra term). Since $n \geq 1$ and $x^2 \geq 0$, we have $nx^2 \geq 0$, so $1 + (n+1)x + nx^2 \geq 1 + (n+1)x$.

Step 7 (chain and conclude). Combining Steps 3–6,

$$(1+x)^{n+1} = (1+x)^n(1+x) \geq (1+nx)(1+x) = 1 + (n+1)x + nx^2 \geq 1 + (n+1)x,$$

which is $P(n+1)$. By induction, $P(n)$ holds for all $n \in \mathbb{N}$. □

Note: the hypothesis $x \geq -1$ is used exactly once, to multiply an inequality by $1+x \geq 0$. Look for where each hypothesis is used.

What goes wrong for $x < -1$? Take $x = -4$, $n = 3$: $(1-4)^3 = -27$ but $1 + 3(-4) = -11$, and $-27 < -11$. So the claim really can fail when $x < -1$, and Step 4 is exactly where the proof breaks.

Example 2.5 (Product identity, HW1 P2 style). For $n \geq 2$: $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$.

Key step: $\frac{n+1}{2n} \cdot \frac{(n+1)^2-1}{(n+1)^2} = \frac{n+1}{2n} \cdot \frac{n(n+2)}{(n+1)^2} = \frac{n+2}{2(n+1)}$, using $(n+1)^2 - 1 = n(n+2)$ (difference of squares).

Full proof, step by step. Let $P(n)$ be the statement $\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \frac{n+1}{2n}$. We prove $P(n)$ for all $n \geq 2$ by induction.

Step 1 (base case $n = 2$). The product has a single factor, $k = 2$:

$$\text{LHS} = 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4}, \quad \text{RHS} = \frac{2+1}{2 \cdot 2} = \frac{3}{4}. \checkmark$$

Step 2 (write the target $P(n+1)$). Replace every n by $n+1$:

$$\prod_{k=2}^{n+1} \left(1 - \frac{1}{k^2}\right) = \frac{(n+1)+1}{2(n+1)} = \frac{n+2}{2(n+1)}.$$

This is what we must reach.

Step 3 (peel off the last factor). Assume $P(n)$. The product up to $n+1$ is the product up to n , times the $k = n+1$ factor:

$$\prod_{k=2}^{n+1} \left(1 - \frac{1}{k^2}\right) = \underbrace{\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right)}_{\text{this is } P(n)} \cdot \left(1 - \frac{1}{(n+1)^2}\right).$$

Step 4 (substitute the inductive hypothesis).

$$= \frac{n+1}{2n} \cdot \left(1 - \frac{1}{(n+1)^2}\right).$$

Step 5 (common denominator in the last factor).

$$1 - \frac{1}{(n+1)^2} = \frac{(n+1)^2}{(n+1)^2} - \frac{1}{(n+1)^2} = \frac{(n+1)^2 - 1}{(n+1)^2}.$$

Step 6 (factor the numerator). Use $a^2 - b^2 = (a-b)(a+b)$ with $a = n+1$, $b = 1$:

$$(n+1)^2 - 1 = ((n+1) - 1)((n+1) + 1) = n(n+2).$$

(Check by expanding: $(n+1)^2 - 1 = n^2 + 2n + 1 - 1 = n^2 + 2n = n(n+2)$.)

Step 7 (multiply and cancel).

$$\frac{n+1}{2n} \cdot \frac{n(n+2)}{(n+1)^2} = \frac{(n+1)n(n+2)}{2n(n+1)^2} = \frac{\cancel{n}(n+1)(n+2)}{2\cancel{n}(n+1)^2} = \frac{n+2}{2(n+1)}.$$

Cancelling is allowed because $n \neq 0$ and $n+1 \neq 0$ (as $n \geq 2$).

Step 8 (conclude). This is exactly the target from Step 2, so $P(n) \Rightarrow P(n+1)$. Together with the base case, $P(n)$ holds for all $n \geq 2$. $\square$

Sanity check $n = 3$: $(1 - \frac{1}{4})(1 - \frac{1}{9}) = \frac{3}{4} \cdot \frac{8}{9} = \frac{24}{36} = \frac{2}{3}$, and $\frac{3+1}{2 \cdot 3} = \frac{4}{6} = \frac{2}{3}$. $\checkmark$

Where does $\frac{n+1}{2n}$ come from? (telescoping) Each factor splits as $1 - \frac{1}{k^2} = \frac{k^2-1}{k^2} = \frac{k-1}{k} \cdot \frac{k+1}{k}$, so

$$\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \left(\frac{1}{2} \cdot \frac{2}{3} \cdots \frac{n-1}{n}\right) \left(\frac{3}{2} \cdot \frac{4}{3} \cdots \frac{n+1}{n}\right) = \frac{1}{n} \cdot \frac{n+1}{2} = \frac{n+1}{2n}.$$

Almost everything cancels in each bracket. This is a good way to *guess* the formula. The induction proof above is what makes it rigorous.

2.5 Contradiction and irrationality (HW1 P4, Week 2)

Method 2.6. Assume the opposite. Solve for the piece you know is irrational. Closure of $\mathbb{Q}$ under $+$, $-$, $\times$, $\div$ then forces that piece to be rational, which is a contradiction.

Example 2.7 ($\sqrt{2} \notin \mathbb{Q}$, from lecture: know this proof cold). Suppose $\sqrt{2} = p/q$ in lowest terms. Then $p^2 = 2q^2$, so p^2 is even. If $p = 2k + 1$ were odd, then $p^2 = 2(2k^2 + 2k) + 1$ would be odd. So p is even, $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ and q is even too. This contradicts lowest terms.

Lemma used twice. For an integer m : if m^2 is even, then m is even.

Proof (contrapositive). Suppose m is odd, so $m = 2k + 1$ for some $k \in \mathbb{Z}$. Then

$$m^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

which is $2 \cdot (\text{integer}) + 1$, so m^2 is odd. Hence “ m odd $\Rightarrow m^2$ odd”, which is equivalent to “ m^2 even $\Rightarrow m$ even”. $\square$

Full proof, step by step. Step 1 (assume the opposite). Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$.

Step 2 (write it in lowest terms). Then $\sqrt{2} = \frac{p}{q}$ with $p, q \in \mathbb{Z}$ and $q \neq 0$. By cancelling common factors we may assume the fraction is in lowest terms, i.e. p and q have *no common factor greater than 1*. (Every fraction can be reduced, for example $\frac{6}{4} = \frac{3}{2}$.) This extra assumption is what we will contradict.

Step 3 (clear the square root). Square both sides and multiply by q^2 :

$$2 = \frac{p^2}{q^2} \implies p^2 = 2q^2.$$

Step 4 (p is even). $p^2 = 2q^2$ is $2 \cdot (\text{integer})$, so p^2 is even. By the Lemma, p is even.

Step 5 (substitute $p = 2k$). Since p is even, $p = 2k$ for some $k \in \mathbb{Z}$. Plug into $p^2 = 2q^2$:

$$(2k)^2 = 2q^2 \implies 4k^2 = 2q^2 \implies q^2 = 2k^2.$$

Step 6 (q is even). $q^2 = 2k^2$ is $2 \cdot (\text{integer})$, so q^2 is even. By the Lemma again, q is even.

Step 7 (contradiction). Now 2 divides both p and q , so they share the common factor 2. This contradicts Step 2 (lowest terms).

Step 8 (conclude). The assumption in Step 1 is false, so $\sqrt{2} \notin \mathbb{Q}$. $\square$

Why lowest terms? Without it there is no contradiction. Finding that p and q are both even would just mean the fraction could be reduced. Lowest terms is the “trap” that the evenness argument springs.

Common point-losers:

- Saying “ p^2 even, so p even” with no justification. Prove the Lemma, or at least cite it.
- Forgetting to say $q \neq 0$ and to assume lowest terms at the start.
- Concluding “ p even” from “ $p^2 = 2q^2$ ” without first noting that p^2 is even.

Sanity check: why doesn’t this “prove” $\sqrt{4} \notin \mathbb{Q}$? Run the same steps: $p^2 = 4q^2$, so p is even, $p = 2k$. Then $4k^2 = 4q^2$, so $q^2 = k^2$. This says *nothing* about q being even, so no contradiction arises, which is correct because $\sqrt{4} = 2 \in \mathbb{Q}$. The factor 2 appearing exactly once in $p^2 = 2q^2$ is what makes Step 6 work.

Same template for $\sqrt{3}$: replace “even/odd” with “divisible by 3 or not.” If $3 \nmid m$, then $m = 3k + 1$ or $m = 3k + 2$, and in both cases m^2 leaves remainder 1 when divided by 3. So $3 \mid m^2 \Rightarrow 3 \mid m$, and the rest of the proof goes through unchanged.

Example 2.8 (rational $\pm$ irrational). Let $r \in \mathbb{Q}$ and $y \notin \mathbb{Q}$. Then

- (a) $r + y \notin \mathbb{Q}$ and $r - y \notin \mathbb{Q}$;
- (b) if also $r \neq 0$, then $ry \notin \mathbb{Q}$ and $y/r \notin \mathbb{Q}$.

Tool: closure of $\mathbb{Q}$. If $a = \frac{p}{q}$ and $b = \frac{s}{t}$ with $p, q, s, t \in \mathbb{Z}$ and $q, t \neq 0$, then

$$a + b = \frac{pt + sq}{qt}, \quad a - b = \frac{pt - sq}{qt}, \quad ab = \frac{ps}{qt}, \quad \frac{a}{b} = \frac{pt}{qs} \quad (\text{if } b \neq 0, \text{ i.e. } s \neq 0).$$

Each is an integer over a nonzero integer, so it is rational. This is the only fact about $\mathbb{Q}$ the proof uses.

Proof of (a), the sum. Step 1 (assume the opposite). Suppose, for contradiction, that $r + y \in \mathbb{Q}$.

Give it a name: $z := r + y$, so $z \in \mathbb{Q}$.

Step 2 (solve for the irrational piece). Subtract r from both sides: $y = z - r$.

Step 3 (apply closure). Both z and r are rational, so $z - r$ is rational by closure under subtraction.

Hence $y \in \mathbb{Q}$.

Step 4 (contradiction). This contradicts the hypothesis $y \notin \mathbb{Q}$. So the assumption in Step 1 was false, and $r + y \notin \mathbb{Q}$. $\square$

Proof of (a), the difference. Suppose $r - y = z \in \mathbb{Q}$. Solving for y gives $y = r - z$, which is rational by closure. This contradicts $y \notin \mathbb{Q}$, so $r - y \notin \mathbb{Q}$. $\square$

Proof of (b), the product. Step 1. Suppose, for contradiction, that $ry \in \mathbb{Q}$. Call it $w := ry$, so $w \in \mathbb{Q}$.

Step 2. Solve for y . Because $r \neq 0$ we may divide by r : $y = \frac{w}{r}$.

Step 3. w and r are rational and $r \neq 0$, so w/r is rational by closure under division. Hence $y \in \mathbb{Q}$.

Step 4. This contradicts $y \notin \mathbb{Q}$, so $ry \notin \mathbb{Q}$.

For the quotient, note $1/r \in \mathbb{Q}$ and $1/r \neq 0$, and $y/r = (1/r) \cdot y$. The product case with $1/r$ in place of r gives $y/r \notin \mathbb{Q}$. $\square$

Where is $r \neq 0$ used? Only in Step 2 of (b), to divide by r . It is really needed: if $r = 0$, then $ry = 0 \in \mathbb{Q}$, so the claim is false.

Concrete check. $3 + \sqrt{2} \notin \mathbb{Q}$: if $3 + \sqrt{2} = z \in \mathbb{Q}$, then $\sqrt{2} = z - 3 \in \mathbb{Q}$, which contradicts the previous example. Likewise $5\sqrt{2} \notin \mathbb{Q}$, because otherwise $\sqrt{2} = \frac{5\sqrt{2}}{5} \in \mathbb{Q}$.

Counterexample question: is irrational + irrational always irrational? No: $\sqrt{2} + (-\sqrt{2}) = 0$. Another one is $\sqrt{2} + (1 - \sqrt{2}) = 1$, where $1 - \sqrt{2} \notin \mathbb{Q}$ by part (a). Is irrational $\times$ irrational always irrational? No: $\sqrt{2} \cdot \sqrt{2} = 2$ and $\sqrt{2} \cdot \sqrt{8} = \sqrt{16} = 4$.

Why the proof above fails here: Step 3 needs r to be rational to conclude $z - r \in \mathbb{Q}$. If r is irrational, closure says nothing, and the argument breaks down.

3 Supremum, infimum, completeness

3.1 Definitions (state these exactly)

Definition 3.1. Let $S \subseteq \mathbb{R}$ be nonempty.

- M is an **upper bound** of S if $s \leq M$ for all $s \in S$.
- $M = \sup S$ if M is an upper bound of S and no $M' < M$ is an upper bound of S . (It is the *least* upper bound.)

- $m = \inf S$ if m is a lower bound of S and no $m' > m$ is a lower bound of S .
- $\max S$ is an upper bound that *belongs to* S . If $\max S$ exists, then $\sup S = \max S$.

Theorem 3.2 (Completeness Axiom). Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$. (It follows that every nonempty set bounded below has an infimum.)

Theorem 3.3 (ε -characterization of sup, the workhorse). $M = \sup S$ if and only if

- (i) $s \leq M$ for all $s \in S$, and
- (ii) for every $\varepsilon > 0$ there is $s \in S$ with $s > M - \varepsilon$.

The same holds for inf with the inequalities flipped: (ii) becomes $\forall \varepsilon > 0 \exists s \in S$ with $s < m + \varepsilon$.

Two facts used constantly:

- If t is *any* upper bound of S , then $\sup S \leq t$. (This is the “least” part.)
- $\sup S - \varepsilon$ is *not* an upper bound, so some element of S exceeds it.

3.2 Archimedean property and density

Theorem 3.4 (Archimedean property). For any $x \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $n > x$. Equivalently: for any $\varepsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \varepsilon$.

Proof idea (contradiction plus completeness): if $\mathbb{N}$ were bounded above, let $M = \sup \mathbb{N}$. Then $M - 1$ is not an upper bound, so some $n > M - 1$. But then $n + 1 > M$ with $n + 1 \in \mathbb{N}$, contradiction.

Theorem 3.5 (Density). If $x < y$, there is $r \in \mathbb{Q}$ with $x < r < y$, and there is an irrational z with $x < z < y$.

Proof idea for $\mathbb{Q}$: pick n with $1/n < y - x$. Let $m = \lfloor nx \rfloor + 1$. Then $nx < m \leq nx + 1 < ny$, so $x < m/n < y$.

Proof idea for irrationals (HW2 P1): either take rationals $x < r_1 < r_2 < y$ and $z = r_1 + \frac{r_2 - r_1}{\sqrt{2}}$, or apply density of $\mathbb{Q}$ to $x/\sqrt{2} < r < y/\sqrt{2}$ with $r \neq 0$ and take $z = r\sqrt{2}$.

3.3 Computing sup and inf of explicit sets

Method 3.6. 1. List the first few elements to guess the sup and inf.

- 2. If the candidate is *attained* (it is an element of S), show it is a bound. It is then the max or min, and you are done.
- 3. If it is *not attained*, show it is a bound, then use Theorem 3.3(ii) together with the Archimedean property.

Example 3.7. $S = \{1 - \frac{1}{n} : n \in \mathbb{N}\} = \{0, \frac{1}{2}, \frac{2}{3}, \dots\}$. Then $\inf S = \min S = 0$ (at $n = 1$) and $\sup S = 1$ (not attained).

Proof that $\sup S = 1$. (i) $1 - \frac{1}{n} < 1$ for all n , so 1 is an upper bound. (ii) Let $\varepsilon > 0$. By the Archimedean property, choose n with $\frac{1}{n} < \varepsilon$. Then $1 - \frac{1}{n} > 1 - \varepsilon$, and $1 - \frac{1}{n} \in S$. By Theorem 3.3, $\sup S = 1$. $\square$

Full proof, step by step. Step 1 ($\inf S = \min S = 0$). For every $n \in \mathbb{N}$, $n \geq 1$, so $\frac{1}{n} \leq 1$ and $1 - \frac{1}{n} \geq 0$. Thus 0 is a lower bound of S . Also $0 = 1 - \frac{1}{1} \in S$. A lower bound that belongs to the set is the minimum, so $\inf S = \min S = 0$.

Step 2 (1 is an upper bound). For every $n \in \mathbb{N}$, $\frac{1}{n} > 0$, so $1 - \frac{1}{n} < 1$. Hence $s \leq 1$ for all $s \in S$. This is part (i) of Theorem 3.3.

Step 3 (start part (ii)). Let $\varepsilon > 0$ be arbitrary. We must find an element of S that is larger than $1 - \varepsilon$.

Step 4 (scratch: what n works?). $1 - \frac{1}{n} > 1 - \varepsilon \iff \frac{1}{n} < \varepsilon$. So we need n with $\frac{1}{n} < \varepsilon$.

Step 5 (Archimedean property). Such an $n \in \mathbb{N}$ exists by the Archimedean property. Fix one.

Step 6 (verify). Then $s := 1 - \frac{1}{n} \in S$ and $s = 1 - \frac{1}{n} > 1 - \varepsilon$.

Step 7 (conclude). Steps 2–6 verify (i) and (ii) of Theorem 3.3, so $\sup S = 1$.

Step 8 (not attained). If $1 \in S$ then $1 = 1 - \frac{1}{n}$ for some n , so $\frac{1}{n} = 0$, which is impossible. So $1 \notin S$ and S has no maximum. $\square$

Example 3.8 (HW2 P2). $S_1 = \{\frac{1}{n}\}$: $\sup = \max = 1$, and $\inf = 0$ is not attained. $S_2 = \{\frac{(-1)^n}{n}\}$: split into odd and even n . Then $\sup S_2 = \max = \frac{1}{2}$ (at $n = 2$) and $\inf S_2 = \min = -1$ (at $n = 1$).

Full proof for S_1 , step by step. *Step 1* ($\sup S_1 = \max S_1 = 1$). For every $n \in \mathbb{N}$, $n \geq 1$, so $\frac{1}{n} \leq 1$. So 1 is an upper bound, and $1 = \frac{1}{1} \in S_1$. An upper bound that belongs to the set is the maximum, so $\sup S_1 = \max S_1 = 1$.

Step 2 (0 is a lower bound). For every n , $\frac{1}{n} > 0$. So $0 \leq s$ for all $s \in S_1$.

Step 3 (the ε part of the inf). Let $\varepsilon > 0$. By the Archimedean property there is $n \in \mathbb{N}$ with $\frac{1}{n} < \varepsilon = 0 + \varepsilon$, and $\frac{1}{n} \in S_1$. By Theorem 3.3 (inf version), $\inf S_1 = 0$.

Step 4 (not attained). $\frac{1}{n} = 0$ has no solution, so $0 \notin S_1$ and S_1 has no minimum. $\square$

Full proof for S_2 , step by step. *Step 1 (list and split).* The elements are $-1, \frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \dots$

- If n is even, then $\frac{(-1)^n}{n} = \frac{1}{n}$, and $n \geq 2$, so $0 < \frac{1}{n} \leq \frac{1}{2}$.

- If n is odd, then $\frac{(-1)^n}{n} = -\frac{1}{n}$, and $n \geq 1$, so $-1 \leq -\frac{1}{n} < 0$.

Step 2 (upper bound $\frac{1}{2}$). Even terms are $\leq \frac{1}{2}$ by Step 1. Odd terms are $< 0 < \frac{1}{2}$. So every element is $\leq \frac{1}{2}$.

Step 3 (attained). $n = 2$ gives $\frac{(-1)^2}{2} = \frac{1}{2} \in S_2$. So $\sup S_2 = \max S_2 = \frac{1}{2}$.

Step 4 (lower bound -1). Odd terms are ≥ -1 by Step 1. Even terms are $> 0 > -1$. So every element is ≥ -1 .

Step 5 (attained). $n = 1$ gives $-1 \in S_2$. So $\inf S_2 = \min S_2 = -1$. $\square$

Lesson: when the candidate is attained, you never need ε . “Bound + element of the set” finishes it.

Example 3.9. $S = (a, b)$: $\sup S = b$ and $\inf S = a$, and neither is attained. For (ii): given $\varepsilon > 0$ (with $\varepsilon < b - a$ if needed), the point $x = b - \varepsilon/2 \in S$ satisfies $x > b - \varepsilon$.

Full proof that $\sup(a, b) = b$ (for $a < b$), step by step. *Step 1 (upper bound).* If $x \in (a, b)$ then $a < x < b$, so $x \leq b$. Thus b is an upper bound.

Step 2 (start part (ii)). Let $\varepsilon > 0$ be arbitrary. We need $x \in (a, b)$ with $x > b - \varepsilon$.

Step 3 (handle large ε). The point $b - \frac{\varepsilon}{2}$ might fall at or below a when ε is large. To avoid this, let $\delta = \min\{\varepsilon, b - a\} > 0$, and set $x = b - \frac{\delta}{2}$.

Step 4 ($x \in (a, b)$). Since $\delta > 0$, $x < b$. Since $\delta \leq b - a$, we have $\frac{\delta}{2} < b - a$, so $x = b - \frac{\delta}{2} > b - (b - a) = a$. So $a < x < b$.

Step 5 ($x > b - \varepsilon$). Since $\delta \leq \varepsilon$, we have $\frac{\delta}{2} < \varepsilon$, so $x = b - \frac{\delta}{2} > b - \varepsilon$.

Step 6 (conclude). By Theorem 3.3, $\sup(a, b) = b$.

Step 7 (not attained). $b \notin (a, b)$ since $b < b$ is false. So (a, b) has no maximum. $\square$

The proof that $\inf(a, b) = a$ is the mirror image: use $x = a + \frac{\delta}{2}$ with the same δ .

3.4 Standard sup/inf proofs: templates

Example 3.10 ($\sup S \leq \inf T$, HW2 P3). If $s \leq t$ for all $s \in S, t \in T$: fix t . Then t is an upper bound of S , so $\sup S \leq t$. Since t was arbitrary, $\sup S$ is a lower bound of T , so $\sup S \leq \inf T$.

Full proof, step by step. Let $S, T \subseteq \mathbb{R}$ be nonempty with $s \leq t$ for all $s \in S, t \in T$.

Step 1 (sup S exists). $T \neq \emptyset$, so pick some $t_0 \in T$. By hypothesis $s \leq t_0$ for all $s \in S$, so S is bounded above. Since $S \neq \emptyset$, the Completeness Axiom gives that $\sup S$ exists. Similarly, any $s_0 \in S$ is a lower bound of T , so $\inf T$ exists.

Step 2 (fix an arbitrary t). Let $t \in T$ be arbitrary.

Step 3 (t is an upper bound of S). By hypothesis $s \leq t$ for every $s \in S$. That is the definition of “ t is an upper bound of S ”.

Step 4 (“least” upper bound). $\sup S$ is less than or equal to *every* upper bound of S . So $\sup S \leq t$.

Step 5 (turn it around). Since $t \in T$ was arbitrary, Step 4 says $\sup S \leq t$ for all $t \in T$. So $\sup S$ is a lower bound of T .

Step 6 (“greatest” lower bound). $\inf T$ is greater than or equal to every lower bound of T . So $\sup S \leq \inf T$. $\square$

Pattern: “fix one, bound, then let it vary” is used twice. You apply the “least” property to S and then the “greatest” property to T .

Example 3.11 ($\sup(A + B) = \sup A + \sup B$, HW2 P4). ($\leq$) Since $a + b \leq \sup A + \sup B$, the right side is an upper bound. ($\geq$) Given $\varepsilon > 0$, pick $a > \sup A - \frac{\varepsilon}{2}$ and $b > \sup B - \frac{\varepsilon}{2}$. Then $a + b > \sup A + \sup B - \varepsilon$. **Splitting ε into halves** is the standard trick.

Full proof, step by step. Let A, B be nonempty and bounded above, and $A + B = \{a + b : a \in A, b \in B\}$. Write $\alpha = \sup A$ and $\beta = \sup B$ (these exist by completeness). We show $\sup(A + B) = \alpha + \beta$ using Theorem 3.3.

Step 1 ($A + B$ is nonempty). Pick $a_0 \in A$ and $b_0 \in B$. Then $a_0 + b_0 \in A + B$.

Step 2 (upper bound). Let $x \in A + B$ be arbitrary. Then $x = a + b$ for some $a \in A, b \in B$. Since $a \leq \alpha$ and $b \leq \beta$, adding gives $x = a + b \leq \alpha + \beta$. So $\alpha + \beta$ is an upper bound of $A + B$. In particular $\sup(A + B)$ exists.

Step 3 (start part (ii)). Let $\varepsilon > 0$. We need some $x \in A + B$ with $x > \alpha + \beta - \varepsilon$.

Step 4 (split ε). $\frac{\varepsilon}{2} > 0$, so $\alpha - \frac{\varepsilon}{2}$ is not an upper bound of A . Hence there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{2}$. In the same way there is $b \in B$ with $b > \beta - \frac{\varepsilon}{2}$.

Step 5 (add). Then $a + b \in A + B$ and

$$a + b > \left(\alpha - \frac{\varepsilon}{2}\right) + \left(\beta - \frac{\varepsilon}{2}\right) = \alpha + \beta - \varepsilon.$$

Step 6 (conclude). Steps 2 and 5 are (i) and (ii) of Theorem 3.3, so $\sup(A + B) = \alpha + \beta$. $\square$

Why $\frac{\varepsilon}{2}$ and not ε ? With $a > \alpha - \varepsilon$ and $b > \beta - \varepsilon$ you only get $a + b > \alpha + \beta - 2\varepsilon$, which is too weak. Each of the two pieces gets half of the budget.

Example 3.12 ($\sup(cA) = c\sup A$ for $c > 0$). Let $\alpha = \sup A$. For $a \in A, ca \leq c\alpha$, so $c\alpha$ is an upper bound of cA . Given $\varepsilon > 0$, pick $a \in A$ with $a > \alpha - \varepsilon/c$. Then $ca > c\alpha - \varepsilon$. (What changes if $c < 0$? Then $\sup(cA) = c\inf A$.)

Full proof, step by step. Let A be nonempty and bounded above, $c > 0$, and $cA = \{ca : a \in A\}$. Let $\alpha = \sup A$.

Step 1 (upper bound). Let $x \in cA$, so $x = ca$ for some $a \in A$. Since $a \leq \alpha$ and $c > 0$, multiplying by c keeps the inequality: $ca \leq c\alpha$. So $c\alpha$ is an upper bound of cA .

Step 2 (start part (ii)). Let $\varepsilon > 0$. We want $x \in cA$ with $x > c\alpha - \varepsilon$.

Step 3 (scratch: rescale ε). If $x = ca$, then $ca > c\alpha - \varepsilon \iff a > \alpha - \frac{\varepsilon}{c}$ (divide by $c > 0$). So we need an element of A within $\frac{\varepsilon}{c}$ of α .

Step 4 (use $\sup A$). $\frac{\varepsilon}{c} > 0$, so $\alpha - \frac{\varepsilon}{c}$ is not an upper bound of A . So there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{c}$.

Step 5 (multiply back). Multiply by $c > 0$: $ca > c\alpha - \varepsilon$, and $ca \in cA$.

Step 6 (conclude). By Theorem 3.3, $\sup(cA) = c\alpha = c \sup A$. $\square$

Where is $c > 0$ used? In Steps 1, 3 and 5, every time we multiply or divide an inequality by c . If $c < 0$ each of those inequalities flips, upper bounds turn into lower bounds, and you get $\sup(cA) = c \inf A$ instead.

Example 3.13 ($\inf S = -\sup(-S)$). $m \leq s$ for all $s \in S \iff -s \leq -m$ for all $s \in S$. So lower bounds of S are exactly the negatives of upper bounds of $-S$, and the greatest lower bound corresponds to the least upper bound.

Full proof, step by step. Let S be nonempty and bounded below, and let $-S = \{-s : s \in S\}$.

Step 1 (key equivalence). For any real m :

$$\begin{aligned} m \text{ is a lower bound of } S &\iff m \leq s \text{ for all } s \in S \\ &\iff -s \leq -m \text{ for all } s \in S \\ &\iff -m \text{ is an upper bound of } -S. \end{aligned}$$

(Multiplying by -1 flips inequalities.)

Step 2 ($\sup(-S)$ exists). S has some lower bound m , so by Step 1 $-m$ is an upper bound of $-S$. Also $-S \neq \emptyset$. By completeness, $\beta := \sup(-S)$ exists.

Step 3 ($-\beta$ is a lower bound of S). β is an upper bound of $-S$, i.e. $-(\beta)$ is an upper bound of $-S$. By Step 1 (with $m = -\beta$), $-\beta$ is a lower bound of S .

Step 4 ($-\beta$ is the greatest). Let m be any lower bound of S . By Step 1, $-m$ is an upper bound of $-S$. Since β is the least upper bound, $\beta \leq -m$, so $m \leq -\beta$.

Step 5 (conclude). $-\beta$ is a lower bound of S and every lower bound is $\leq -\beta$. By definition, $\inf S = -\beta = -\sup(-S)$. $\square$

Use: this lets you get every inf fact from the matching sup fact, and also shows that completeness for sups gives completeness for infs.

3.5 Monotonicity of sup

If $A \subseteq B$, then $\sup A \leq \sup B$ and $\inf A \geq \inf B$. (Any upper bound of B is an upper bound of A .)

4 Absolute value

Theorem 4.1 (Properties).

1. $|xy| = |x||y|$ and $|-x| = |x|$.
2. For $a \geq 0$: $|x| \leq a \iff -a \leq x \leq a$. **(Most useful fact.)**

3. Triangle inequality: $|x + y| \leq |x| + |y|$.
4. Reverse triangle inequality: $||x| - |y|| \leq |x - y|$.
5. $|x - a| < \delta \iff a - \delta < x < a + \delta$.

Method 4.2 (Proving the triangle inequality). Add $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ to get $-(|x| + |y|) \leq x + y \leq |x| + |y|$, then apply property 2.

Method 4.3 (Reverse triangle inequality, HW3 P1). Write $x = (x - y) + y$, so $|x| \leq |x - y| + |y|$, i.e. $|x| - |y| \leq |x - y|$. Swap x and y to get $-(|x| - |y|) \leq |x - y|$. Then property 2 gives the result.

Method 4.4 (The add-and-subtract trick). To compare a and b through c , write $a - b = (a - c) + (c - b)$ and apply the triangle inequality. This is how uniqueness of limits and the product rule are proved.

Example 4.5 (Solving equations with absolute values). Solve $|x - 3| + |x - 1| = 3$. Break the line at the points 1 and 3:

- $x < 1$: $(3 - x) + (1 - x) = 3 \Rightarrow x = \frac{1}{2}$. ✓
- $1 \leq x < 3$: $(3 - x) + (x - 1) = 2 \neq 3$. No solution.
- $x \geq 3$: $(x - 3) + (x - 1) = 3 \Rightarrow x = \frac{7}{2}$. ✓

Full solution, step by step. Step 1 (find the break points). $|x - 3|$ changes formula at $x = 3$, and $|x - 1|$ changes formula at $x = 1$. These points split $\mathbb{R}$ into $(-\infty, 1)$, $[1, 3)$ and $[3, \infty)$.

Step 2 (sign table). Use $|u| = u$ if $u \geq 0$ and $|u| = -u$ if $u < 0$:

	$x - 3$	$x - 1$
$x < 1$	< 0 , so $ x - 3 = 3 - x$	< 0 , so $ x - 1 = 1 - x$
$1 \leq x < 3$	< 0 , so $ x - 3 = 3 - x$	≥ 0 , so $ x - 1 = x - 1$
$x \geq 3$	≥ 0 , so $ x - 3 = x - 3$	> 0 , so $ x - 1 = x - 1$

Step 3 (case $x < 1$). The equation becomes $(3 - x) + (1 - x) = 3$, i.e. $4 - 2x = 3$, so $x = \frac{1}{2}$. Check that it lies in the case: $\frac{1}{2} < 1$. ✓

Step 4 (case $1 \leq x < 3$). The equation becomes $(3 - x) + (x - 1) = 3$, i.e. $2 = 3$. This is false, so there is no solution in this interval.

Step 5 (case $x \geq 3$). The equation becomes $(x - 3) + (x - 1) = 3$, i.e. $2x - 4 = 3$, so $x = \frac{7}{2}$. Check: $\frac{7}{2} \geq 3$. ✓

Step 6 (verify in the original equation). $|\frac{1}{2} - 3| + |\frac{1}{2} - 1| = \frac{5}{2} + \frac{1}{2} = 3$ and $|\frac{7}{2} - 3| + |\frac{7}{2} - 1| = \frac{1}{2} + \frac{5}{2} = 3$.

Step 7 (answer). The solution set is $\{\frac{1}{2}, \frac{7}{2}\}$. $\square$

Always check that a case's answer lies in that case's interval. If it doesn't, throw it away.

Geometric check: $|x - 3| + |x - 1|$ is the total distance from x to 1 and to 3. Between 1 and 3 this total is always 2. You must move $\frac{1}{2}$ outside the interval on either side to get 3.

5 Limits of sequences: the ε - N definition

Definition 5.1. $x_n \rightarrow L$ means: for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n - L| < \varepsilon$ for all $n \geq N$.

Method 5.2 (The ε - N recipe). 1. **Guess** L (leading terms, or intuition from calculus).

2. **Scratch work:** simplify $|x_n - L|$ to a single fraction.

3. **Overestimate** it by something simpler, such as $\frac{C}{n}$. You make a fraction larger by making the denominator smaller or the numerator larger. You only need an upper bound.

4. Solve $\frac{C}{n} < \varepsilon$, which gives $n > \frac{C}{\varepsilon}$, so $N > \frac{C}{\varepsilon}$ (Archimedean).

5. **Write it forwards:** “Let $\varepsilon > 0$. Choose $N > \dots$. Then for $n \geq N$: $|x_n - L| = \dots \leq \dots < \varepsilon$.”

Example 5.3 (HW3 P2). $\frac{n+2}{3n+5} \rightarrow \frac{1}{3}$.

Scratch work (not part of the proof): $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{3(3n+5)} = \frac{1}{9n+15} < \frac{1}{9n}$. We want $\frac{1}{9n} < \varepsilon$, i.e. $n > \frac{1}{9\varepsilon}$.

Proof. Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{9\varepsilon}$. For $n \geq N$, $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n} \leq \frac{1}{9N} < \varepsilon$. $\square$

Full proof, step by step. Step 1 (simplify $|x_n - L|$, scratch). Use the common denominator $3(3n+5)$:

$$\frac{n+2}{3n+5} - \frac{1}{3} = \frac{3(n+2) - (3n+5)}{3(3n+5)} = \frac{3n+6-3n-5}{9n+15} = \frac{1}{9n+15}.$$

This is positive, so $\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15}$.

Step 2 (overestimate, scratch). Since $9n+15 > 9n > 0$, $\frac{1}{9n+15} < \frac{1}{9n}$. (A smaller denominator gives a larger fraction.)

Step 3 (solve for n , scratch). $\frac{1}{9n} < \varepsilon \iff 9n\varepsilon > 1 \iff n > \frac{1}{9\varepsilon}$.

Step 4 (the proof starts here: let $\varepsilon > 0$). Let $\varepsilon > 0$ be arbitrary.

Step 5 (choose N). By the Archimedean property, choose $N \in \mathbb{N}$ with $N > \frac{1}{9\varepsilon}$. Note that N depends only on ε .

Step 6 (verify for every $n \geq N$). Let $n \geq N$. Then $9n \geq 9N > \frac{1}{\varepsilon}$, so $\frac{1}{9n} \leq \frac{1}{9N} < \varepsilon$. Therefore

$$\left| \frac{n+2}{3n+5} - \frac{1}{3} \right| = \frac{1}{9n+15} < \frac{1}{9n} \leq \frac{1}{9N} < \varepsilon.$$

Step 7 (conclude). Since $\varepsilon > 0$ was arbitrary, $\frac{n+2}{3n+5} \rightarrow \frac{1}{3}$ by definition. $\square$

Example 5.4 (Lecture). $\frac{2n-1}{n+1} \rightarrow 2$: $\left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1} < \frac{3}{n}$. Choose $N > \frac{3}{\varepsilon}$.

Full proof, step by step. Step 1 (simplify, scratch).

$$\frac{2n-1}{n+1} - 2 = \frac{2n-1-2(n+1)}{n+1} = \frac{-3}{n+1}, \quad \text{so} \quad \left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1}.$$

Step 2 (overestimate, scratch). $n+1 > n$, so $\frac{3}{n+1} < \frac{3}{n}$. Then $\frac{3}{n} < \varepsilon \iff n > \frac{3}{\varepsilon}$.

Step 3 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \frac{3}{\varepsilon}$.

Step 4 (verify). For $n \geq N$:

$$\left| \frac{2n-1}{n+1} - 2 \right| = \frac{3}{n+1} < \frac{3}{n} \leq \frac{3}{N} < \varepsilon.$$

Step 5 (conclude). So $\frac{2n-1}{n+1} \rightarrow 2$. $\square$

Watch the sign: the difference is $\frac{-3}{n+1}$, which is negative. The absolute value removes the minus sign, so don't skip it.

Example 5.5 (A numerator that needs bounding). $\frac{3n^2+n}{n^2+1} \rightarrow 3$: $\left| \frac{3n^2+n}{n^2+1} - 3 \right| = \frac{|n-3|}{n^2+1} \leq \frac{n+3}{n^2} \leq \frac{4n}{n^2} = \frac{4}{n}$ for $n \geq 1$ (since $3 \leq 3n$). Choose $N > \frac{4}{\varepsilon}$. **Bound the numerator above and the denominator below.**

Full proof, step by step. Step 1 (simplify, scratch).

$$\frac{3n^2+n}{n^2+1} - 3 = \frac{3n^2+n-3n^2-3}{n^2+1} = \frac{n-3}{n^2+1}, \quad \text{so} \quad \left| \frac{3n^2+n}{n^2+1} - 3 \right| = \frac{|n-3|}{n^2+1}.$$

Step 2 (numerator: bound above). By the triangle inequality, $|n-3| \leq |n| + |-3| = n+3$. Since $n \geq 1$, $3 \leq 3n$, so $n+3 \leq 4n$.

Step 3 (denominator: bound below). $n^2+1 > n^2$, so $\frac{1}{n^2+1} < \frac{1}{n^2}$.

Step 4 (combine).

$$\frac{|n-3|}{n^2+1} < \frac{4n}{n^2} = \frac{4}{n}.$$

(For $n=3$ the left side is 0, which is still $< \frac{4}{n}$.)

Step 5 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{4}{\varepsilon}$.

Step 6 (verify). For $n \geq N$: $\left| \frac{3n^2+n}{n^2+1} - 3 \right| < \frac{4}{n} \leq \frac{4}{N} < \varepsilon$.

Step 7 (conclude). So $\frac{3n^2+n}{n^2+1} \rightarrow 3$. □

Why not solve $\frac{|n-3|}{n^2+1} < \varepsilon$ exactly? You could, but it needs the quadratic formula and cases. Overestimating by $\frac{4}{n}$ is valid and much faster. You only need *some* N that works, not the smallest one.

Example 5.6 (Conjugate trick, lecture). $\frac{\sqrt{n^2+n}}{n} \rightarrow 1$: $\frac{\sqrt{n^2+n}-n}{n} = \frac{n}{n(\sqrt{n^2+n}+n)} = \frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n}$. Choose $N > \frac{1}{\varepsilon}$. **Use $\sqrt{a}-\sqrt{b} = \frac{a-b}{\sqrt{a}+\sqrt{b}}$.**

Full proof, step by step. Step 1 (common denominator, scratch). $\frac{\sqrt{n^2+n}}{n} - 1 = \frac{\sqrt{n^2+n}-n}{n}$.

Step 2 (multiply by the conjugate).

$$\sqrt{n^2+n}-n = \frac{(\sqrt{n^2+n}-n)(\sqrt{n^2+n}+n)}{\sqrt{n^2+n}+n} = \frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n} = \frac{n}{\sqrt{n^2+n}+n}.$$

Step 3 (substitute and cancel).

$$\frac{\sqrt{n^2+n}-n}{n} = \frac{n}{n(\sqrt{n^2+n}+n)} = \frac{1}{\sqrt{n^2+n}+n}.$$

This is positive, so it equals its absolute value.

Step 4 (overestimate). $\sqrt{n^2+n} > 0$, so $\sqrt{n^2+n}+n > n$ and $\frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n}$.

Step 5 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$.

Step 6 (verify). For $n \geq N$:

$$\left| \frac{\sqrt{n^2+n}}{n} - 1 \right| = \frac{1}{\sqrt{n^2+n}+n} < \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Step 7 (conclude). So $\frac{\sqrt{n^2+n}}{n} \rightarrow 1$. □

Why the conjugate? $\sqrt{n^2 + n} - n$ is a difference of two large, nearly equal numbers, so you can't see its size. After multiplying by the conjugate, the n^2 terms cancel.

Example 5.7 (Bounded over growing). $\frac{\sin n}{n} \rightarrow 0$: $|\frac{\sin n}{n}| \leq \frac{1}{n} < \varepsilon$ when $n > \frac{1}{\varepsilon}$. (You cannot use the quotient rule, because $\lim \sin n$ does not exist.)

Full proof, step by step. Step 1 (the only fact about $\sin$). $|\sin t| \leq 1$ for all real t . In particular $|\sin n| \leq 1$.

Step 2 (bound the term). For $n \in \mathbb{N}$: $|\frac{\sin n}{n} - 0| = \frac{|\sin n|}{n} \leq \frac{1}{n}$.

Step 3 (let $\varepsilon > 0$ and choose N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$.

Step 4 (verify). For $n \geq N$: $|\frac{\sin n}{n} - 0| \leq \frac{1}{n} \leq \frac{1}{N} < \varepsilon$.

Step 5 (conclude). So $\frac{\sin n}{n} \rightarrow 0$. □

Example 5.8 (Proving divergence). $x_n = (-1)^n$ has no limit. Suppose $x_n \rightarrow L$. Take $\varepsilon = \frac{1}{2}$ and get N . Choose an even $n \geq N$ and an odd $m \geq N$. Then $|1 - L| < \frac{1}{2}$ and $|-1 - L| < \frac{1}{2}$, so $2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 1$, which is a contradiction.

Full proof, step by step. Step 1 (assume the opposite). Suppose, for contradiction, that $(-1)^n \rightarrow L$ for some $L \in \mathbb{R}$.

Step 2 (choose one specific ε). The definition holds for every $\varepsilon > 0$, so it holds for $\varepsilon = \frac{1}{2}$. So there is $N \in \mathbb{N}$ with $|(-1)^n - L| < \frac{1}{2}$ for all $n \geq N$.

Step 3 (pick one even and one odd index past N). Take $n = 2N$ (even) and $m = 2N + 1$ (odd). Both are $\geq N$.

Step 4 (read off the two inequalities). $(-1)^{2N} = 1$ and $(-1)^{2N+1} = -1$, so

$$|1 - L| < \frac{1}{2} \quad \text{and} \quad |-1 - L| < \frac{1}{2}.$$

Step 5 (add and subtract L , then triangle inequality).

$$2 = |1 - (-1)| = |(1 - L) + (L - (-1))| \leq |1 - L| + |L + 1| < \frac{1}{2} + \frac{1}{2} = 1.$$

(Here $|L + 1| = |-1 - L|$.)

Step 6 (contradiction). Step 5 says $2 < 1$, which is false. So no such L exists, and $(-1)^n$ diverges. □

Why $\varepsilon = \frac{1}{2}$? The terms 1 and -1 are 2 apart. Any $\varepsilon \leq 1$ works, because two points within ε of L are less than $2\varepsilon \leq 2$ apart. $\varepsilon = \frac{1}{2}$ leaves some room.

5.1 Limits equal to $\pm\infty$

Definition 5.9. $x_n \rightarrow \infty$ means: for every M there is N such that $x_n > M$ for all $n \geq N$. Also, $x_n \rightarrow -\infty$ means $-x_n \rightarrow \infty$.

Method 5.10. Find a *simpler lower bound* $x_n \geq g(n)$ with $g(n) \rightarrow \infty$ obviously, such as $g(n) = cn$. Then choose N so that $g(n) > M$.

Example 5.11 (Lecture). $\frac{n^4 - 4n}{n^2} \rightarrow \infty$: for $n \geq 2$, $n^4 - 4n \geq n^3$ (since $n^3(n - 1) \geq n^3 \geq 4n$ when $n \geq 2$), so $\frac{n^4 - 4n}{n^2} \geq n$. Given M , choose $N > \max(M, 2)$.

Full proof, step by step. Step 1 (simplify). $\frac{n^4-4n}{n^2} = n^2 - \frac{4}{n}$. We'll work with the original form, since the lower bound is easy to see there.

Step 2 (lower bound on the numerator, for $n \geq 2$). We claim $n^4 - 4n \geq n^3$, i.e. $n^4 - n^3 \geq 4n$. Factor: $n^4 - n^3 = n^3(n - 1)$. For $n \geq 2$ we have $n - 1 \geq 1$, so $n^3(n - 1) \geq n^3 = n \cdot n^2 \geq n \cdot 4 = 4n$ (since $n^2 \geq 4$).

Step 3 (simpler lower bound for x_n). Dividing Step 2 by $n^2 > 0$: for $n \geq 2$, $\frac{n^4-4n}{n^2} \geq \frac{n^3}{n^2} = n$.

Step 4 (let M be arbitrary and choose N). Let $M \in \mathbb{R}$. By the Archimedean property choose $N \in \mathbb{N}$ with $N > \max(M, 2)$.

Step 5 (verify). Let $n \geq N$. Then $n \geq 2$, so Step 3 applies, and $\frac{n^4-4n}{n^2} \geq n \geq N > M$.

Step 6 (conclude). For every M there is such an N , so $\frac{n^4-4n}{n^2} \rightarrow \infty$. $\square$

Why the max? $N > M$ gives the size condition, and $N \geq 2$ makes Step 3 valid. Taking the max gets both at once.

Example 5.12 (HW3 P3: $\infty + \text{bounded} = \infty$). Say $|y_n| \leq B$. Then $x_n + y_n \geq x_n - B$. Given K , apply $x_n \rightarrow \infty$ with the **threshold $K + B$** to get N . For $n \geq N$, $x_n + y_n > K + B - B = K$.

Full proof, step by step. Assume $x_n \rightarrow \infty$ and (y_n) is bounded. We show $x_n + y_n \rightarrow \infty$.

Step 1 (name the bound). Since (y_n) is bounded, there is $B > 0$ with $|y_n| \leq B$ for all n . In particular $-B \leq y_n$ for all n .

Step 2 (lower bound for the sum). Adding x_n gives $x_n + y_n \geq x_n - B$ for all n .

Step 3 (let K be arbitrary). Let $K \in \mathbb{R}$. We must find N with $x_n + y_n > K$ for all $n \geq N$.

Step 4 (scratch: what do we need from x_n ?). By Step 2, it is enough that $x_n - B > K$, i.e. $x_n > K + B$.

Step 5 (apply $x_n \rightarrow \infty$ with threshold $K + B$). The definition of $x_n \rightarrow \infty$ holds for every real threshold, in particular for $M = K + B$. So there is N with $x_n > K + B$ for all $n \geq N$.

Step 6 (verify). For $n \geq N$: $x_n + y_n \geq x_n - B > (K + B) - B = K$.

Step 7 (conclude). K was arbitrary, so $x_n + y_n \rightarrow \infty$. $\square$

Only the lower bound $y_n \geq -B$ is used. So the result holds whenever (y_n) is merely bounded below.

6 Limit theorems (Week 3–4)

Theorem 6.1. Suppose $x_n \rightarrow x$ and $y_n \rightarrow y$. Then:

1. **Uniqueness:** limits are unique.
2. **Bounded:** a convergent sequence is bounded.
3. $ax_n + by_n \rightarrow ax + by$.
4. $x_n y_n \rightarrow xy$.
5. $x_n / y_n \rightarrow x / y$ if $y \neq 0$.
6. **Squeeze:** if $a_n \leq z_n \leq b_n$ and $a_n, b_n \rightarrow L$, then $z_n \rightarrow L$.
7. **Order:** if $x_n \leq y_n$ for all n (large), then $x \leq y$. In particular, $x_n \geq 0 \Rightarrow x \geq 0$. **Strict inequality is not preserved:** $\frac{1}{n} > 0$ but the limit is 0.

Proofs to know (short ones are quiz favorites):

- *Uniqueness:* $|x - x'| \leq |x - x_n| + |x_n - x'| < 2\varepsilon$ for every ε , so $x = x'$.
- *Bounded:* with $\varepsilon = 1$, $|x_n| < |x| + 1$ for $n \geq N$. Take $M = \max\{|x_1|, \dots, |x_{N-1}|, |x| + 1\}$.

- *Sum:* $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$, using $N = \max(N_1, N_2)$.
- *Product:* $|x_n y_n - xy| = |(x_n - x)y_n + x(y_n - y)| \leq M|x_n - x| + |x||y_n - y|$, where $|y_n| \leq M$ because convergent sequences are bounded.
- *Squeeze:* $L - \varepsilon < a_n \leq z_n \leq b_n < L + \varepsilon$.
- *Order, by contradiction:* if $x_n \geq 0$ and $x < 0$, take $\varepsilon = -x/2$. Then $x_n < x + \varepsilon = x/2 < 0$ for large n , contradiction.

6.1 The quotient rule and the “bounded away from zero” lemma

Theorem 6.2 (Lemma). If $y_n \rightarrow y \neq 0$, then there is N_0 such that $|y_n| > \frac{|y|}{2}$ for all $n \geq N_0$.

Proof. Apply the definition with $\varepsilon = \frac{|y|}{2}$. Then $|y_n| \geq |y| - |y - y_n| > |y| - \frac{|y|}{2} = \frac{|y|}{2}$ by the reverse triangle inequality. $\square$

Quotient rule sketch: by the product rule, it suffices to show $\frac{1}{y_n} \rightarrow \frac{1}{y}$. For $n \geq N_0$,

$$\left| \frac{1}{y_n} - \frac{1}{y} \right| = \frac{|y - y_n|}{|y_n||y|} < \frac{2}{|y|^2} |y_n - y|,$$

which can be made $< \varepsilon$ by choosing N with $|y_n - y| < \frac{|y|^2 \varepsilon}{2}$.

6.2 More limit tools

- **Zero times bounded:** if $x_n \rightarrow 0$ and $|y_n| \leq M$, then $x_n y_n \rightarrow 0$, because $|x_n y_n| \leq M|x_n|$. (This does *not* need y_n to converge.)
- **Shifts don’t matter:** $x_n \rightarrow L \iff x_{n+1} \rightarrow L \iff x_{n+k} \rightarrow L$. Changing finitely many terms doesn’t change the limit.
- **Reciprocals and infinity:** if $x_n \rightarrow \infty$, then $\frac{1}{x_n} \rightarrow 0$. If $x_n > 0$ and $x_n \rightarrow 0$, then $\frac{1}{x_n} \rightarrow \infty$.
- **Infinity arithmetic:** $x_n \rightarrow \infty$ and $y_n \geq c > 0$ eventually $\Rightarrow x_n y_n \rightarrow \infty$. $x_n \rightarrow \infty$ and y_n bounded below $\Rightarrow x_n + y_n \rightarrow \infty$. $\infty - \infty$, $0 \cdot \infty$ and $\frac{\infty}{\infty}$ are **indeterminate**: for example $n - n = 0$, $(n + 1) - n = 1$, and $n^2 - n \rightarrow \infty$.
- **Comparison for infinity:** if $x_n \leq y_n$ and $x_n \rightarrow \infty$, then $y_n \rightarrow \infty$.
- **The “max of the N ’s”:** when you need two conditions at once, take $N = \max(N_1, N_2)$.
- **ε -splitting:** to get a total of $< \varepsilon$ from two pieces, make each piece $< \frac{\varepsilon}{2}$. With a constant C in front, make the piece $< \frac{\varepsilon}{C}$.

6.3 Binomial-theorem estimates (Week 4 style)

For $x \geq 0$ and $n \geq 2$: $(1 + x)^n = \sum_{k=0}^n \binom{n}{k} x^k \geq 1 + nx$ and also $(1 + x)^n \geq \binom{n}{2} x^2 = \frac{n(n-1)}{2} x^2$. Keep only the one term you need.

Example 6.3 ($\frac{n}{2^n} \rightarrow 0$). $2^n = (1 + 1)^n \geq \binom{n}{2} = \frac{n(n-1)}{2}$, so for $n \geq 2$, $0 < \frac{n}{2^n} \leq \frac{2n}{n(n-1)} = \frac{2}{n-1} \rightarrow 0$. Squeeze.

Full proof, step by step. Step 1 (binomial theorem). For $n \geq 2$,

$$2^n = (1+1)^n = \sum_{k=0}^n \binom{n}{k} \geq \binom{n}{2} = \frac{n(n-1)}{2},$$

because every term of the sum is ≥ 0 and we kept only the $k = 2$ term.

Step 2 (bound the sequence). For $n \geq 2$, $n-1 > 0$, so taking reciprocals (both sides positive) gives $\frac{1}{2^n} \leq \frac{2}{n(n-1)}$. Multiplying by $n > 0$:

$$0 < \frac{n}{2^n} \leq \frac{2n}{n(n-1)} = \frac{2}{n-1}.$$

Step 3 ($\frac{2}{n-1} \rightarrow 0$). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > 1 + \frac{2}{\varepsilon}$ (also $N \geq 2$). For $n \geq N$, $n-1 \geq N-1 > \frac{2}{\varepsilon}$, so $0 < \frac{2}{n-1} < \varepsilon$.

Step 4 (squeeze). $0 \leq \frac{n}{2^n} \leq \frac{2}{n-1}$ for $n \geq 2$, and both outer sequences tend to 0. By the Squeeze Theorem, $\frac{n}{2^n} \rightarrow 0$. (Or directly: for $n \geq N$, $|\frac{n}{2^n}| \leq \frac{2}{n-1} < \varepsilon$.) $\square$

Why $\binom{n}{2}$ and not $1+n$? Bernoulli gives $2^n \geq 1+n$, so $\frac{n}{2^n} \leq \frac{n}{n+1}$, which does *not* go to 0. You need a term of degree 2 in n to beat the n on top.

Example 6.4 ($a^{1/n} \rightarrow 1$ for $a > 1$). Let $d_n = a^{1/n} - 1 > 0$. Then $a = (1+d_n)^n \geq 1+nd_n$, so $0 < d_n \leq \frac{a-1}{n} \rightarrow 0$. For $0 < a < 1$, apply this to $\frac{1}{a}$ and take reciprocals.

Full proof, step by step. Step 1 (define the error term). Let $a > 1$ and $d_n := a^{1/n} - 1$. Since $a > 1$, $a^{1/n} > 1$ (if $a^{1/n} \leq 1$ then $a = (a^{1/n})^n \leq 1$). So $d_n > 0$.

Step 2 (rewrite a). $a^{1/n} = 1 + d_n$, so $a = (1+d_n)^n$.

Step 3 (Bernoulli). Since $d_n > 0 \geq -1$, Bernoulli's inequality gives $(1+d_n)^n \geq 1+nd_n$. So $a \geq 1+nd_n$.

Step 4 (solve for d_n). $nd_n \leq a-1$, so $0 < d_n \leq \frac{a-1}{n}$.

Step 5 (ε - N). Let $\varepsilon > 0$ and choose $N > \frac{a-1}{\varepsilon}$. For $n \geq N$: $|a^{1/n} - 1| = d_n \leq \frac{a-1}{n} \leq \frac{a-1}{N} < \varepsilon$. So $a^{1/n} \rightarrow 1$.

Step 6 (the case $a = 1$). $1^{1/n} = 1$ for all n , so the limit is 1.

Step 7 (the case $0 < a < 1$). Then $b := \frac{1}{a} > 1$, so $b^{1/n} \rightarrow 1$ by Step 5. Since $a^{1/n} = \frac{1}{b^{1/n}}$ and the limit $1 \neq 0$, the quotient rule gives $a^{1/n} \rightarrow \frac{1}{1} = 1$. $\square$

Example 6.5 ($n^{1/n} \rightarrow 1$). Let $d_n = n^{1/n} - 1 \geq 0$. Then $n = (1+d_n)^n \geq \frac{n(n-1)}{2}d_n^2$, so $d_n^2 \leq \frac{2}{n-1}$ and $0 \leq d_n \leq \sqrt{\frac{2}{n-1}} \rightarrow 0$.

Full proof, step by step. Step 1 (error term). Let $d_n := n^{1/n} - 1$. Since $n \geq 1$, $n^{1/n} \geq 1$ (as in Step 1 above), so $d_n \geq 0$.

Step 2 (binomial theorem, keep the $k = 2$ term). For $n \geq 2$,

$$n = (1+d_n)^n = \sum_{k=0}^n \binom{n}{k} d_n^k \geq \binom{n}{2} d_n^2 = \frac{n(n-1)}{2} d_n^2,$$

since every term is ≥ 0 (here $d_n \geq 0$ is used).

Step 3 (solve for d_n). Divide by $\frac{n(n-1)}{2} > 0$: $d_n^2 \leq \frac{2n}{n(n-1)} = \frac{2}{n-1}$. Since $d_n \geq 0$, taking square roots gives $0 \leq d_n \leq \sqrt{\frac{2}{n-1}}$.

Step 4 (scratch). $\sqrt{\frac{2}{n-1}} < \varepsilon \iff \frac{2}{n-1} < \varepsilon^2 \iff n > 1 + \frac{2}{\varepsilon^2}.$

Step 5 (ε - N). Let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $N > 1 + \frac{2}{\varepsilon^2}$ (so also $N \geq 2$). For $n \geq N$:

$$|n^{1/n} - 1| = d_n \leq \sqrt{\frac{2}{n-1}} \leq \sqrt{\frac{2}{N-1}} < \varepsilon.$$

Step 6 (conclude). So $n^{1/n} \rightarrow 1$. □

Why not Bernoulli here? It gives $n \geq 1 + nd_n$, i.e. $d_n \leq \frac{n-1}{n}$, which does not go to 0. The $\binom{n}{2}$ term is needed because the base n grows.

Example 6.6 ($\frac{a^n}{n!} \rightarrow 0$ for any a). Fix an integer $K \geq 2|a|$. For $n > K$, $\frac{|a|^n}{n!} = \frac{|a|^K}{K!} \cdot \prod_{j=K+1}^n \frac{|a|}{j} \leq \frac{|a|^K}{K!} \left(\frac{1}{2}\right)^{n-K} \rightarrow 0$.

Full proof, step by step. *Step 1 (reduce to $|a|$).* $\left|\frac{a^n}{n!} - 0\right| = \frac{|a|^n}{n!}$, so it is enough to show $\frac{|a|^n}{n!} \rightarrow 0$. If $a = 0$ every term is 0, so assume $a \neq 0$.

Step 2 (fix a cutoff K). By the Archimedean property choose an integer $K \geq 1$ with $K \geq 2|a|$. Then for every $j \geq K+1$, $j > 2|a|$, so $\frac{|a|}{j} < \frac{1}{2}$.

Step 3 (split the product). For $n > K$, write $|a|^n = |a|^K \cdot |a|^{n-K}$ and $n! = K! \cdot (K+1)(K+2) \cdots n$. So

$$\frac{|a|^n}{n!} = \frac{|a|^K}{K!} \cdot \frac{|a|}{K+1} \cdot \frac{|a|}{K+2} \cdots \frac{|a|}{n}.$$

There are exactly $n - K$ factors after the first.

Step 4 (bound each tail factor). By Step 2 each of those $n - K$ factors is $< \frac{1}{2}$. Let $C := \frac{|a|^K}{K!}$, a constant (it does not depend on n). Then

$$0 < \frac{|a|^n}{n!} \leq C \left(\frac{1}{2}\right)^{n-K} = C 2^K \left(\frac{1}{2}\right)^n.$$

Step 5 (the bound tends to 0). $\left(\frac{1}{2}\right)^n \rightarrow 0$ (basic limit, $|\frac{1}{2}| < 1$), so $C 2^K \left(\frac{1}{2}\right)^n \rightarrow 0$ by the constant multiple rule.

Step 6 (squeeze). $0 < \frac{|a|^n}{n!} \leq C 2^K \left(\frac{1}{2}\right)^n$ for $n > K$. By the Squeeze Theorem (only large n matter), $\frac{|a|^n}{n!} \rightarrow 0$, and therefore $\frac{a^n}{n!} \rightarrow 0$. □

Idea: the first K factors can be large, but there are only finitely many of them, so they form a fixed constant C . After that, each new factor at least halves the term.

6.4 Basic limits to memorize

1. $\frac{1}{n^p} \rightarrow 0$ for $p > 0$.
2. **Ratio of polynomials of equal degree:** the limit is the ratio of leading coefficients. (Divide the top and bottom by n^k .) If the top has lower degree, the limit is 0. If the top has higher degree, the limit is $\pm\infty$.
3. $a^n \rightarrow 0$ if $|a| < 1$. (Proof: write $\frac{1}{|a|} = 1 + x$ and use Bernoulli: $|a|^n \leq \frac{1}{1+nx} \rightarrow 0$.)
4. $1 + a + \cdots + a^n = \frac{1-a^{n+1}}{1-a} \rightarrow \frac{1}{1-a}$ if $|a| < 1$.
5. $\frac{\text{bounded}}{\rightarrow \infty} \rightarrow 0$, e.g. $\frac{\sin n}{n}$ and $\frac{(-1)^n}{n}$.
6. If $x_n \rightarrow L$ then $|x_n| \rightarrow |L|$ (reverse triangle inequality).

7 Monotone sequences and the Monotone Convergence Theorem

Definition 7.1. (x_n) is **increasing** if $x_n \leq x_{n+1}$ for all n , **decreasing** if $x_n \geq x_{n+1}$ for all n , and **monotone** if it is one of the two.

Method 7.2 (Showing monotonicity: three tools). 1. Show $x_{n+1} - x_n \geq 0$ (HW3 P4: common denominator, then check the sign).
2. If $x_n > 0$, show $\frac{x_{n+1}}{x_n} \geq 1$.
3. For recursive sequences, use induction: $x_n \leq x_{n+1} \Rightarrow x_{n+1} \leq x_{n+2}$.

Theorem 7.3 (Monotone Convergence Theorem). A bounded monotone sequence converges. If it is increasing and bounded above, then $x_n \rightarrow \sup\{x_n\}$.

Proof. Let $L = \sup\{x_n\}$, which exists by completeness. Let $\varepsilon > 0$. Since $L - \varepsilon$ is not an upper bound, there is N with $x_N > L - \varepsilon$. For $n \geq N$: $L - \varepsilon < x_N \leq x_n \leq L < L + \varepsilon$, so $|x_n - L| < \varepsilon$. $\square$

This proof is short and is exactly Theorem 3.3(ii) plus monotonicity. Be able to reproduce it.

Method 7.4 (Recursive sequences: the 4-step routine). 1. Compute x_1, x_2, x_3 to guess the direction and the limit.
2. Prove a bound by induction.
3. Prove monotonicity (often by induction, or using the bound).
4. By MCT, $x_n \rightarrow L$. Pass to the limit in the recursion using the limit laws, and *also* $x_{n+1} \rightarrow L$ (it is the same sequence shifted). Solve for L and use the bounds to discard wrong roots.

Example 7.5. $x_1 = 1$, $x_{n+1} = \sqrt{2 + x_n}$. Then $x_n \rightarrow 2$.

Proof. Bounded: $0 < x_n < 2$ by induction. Indeed $x_1 = 1$, and if $0 < x_n < 2$ then $0 < x_{n+1} = \sqrt{2 + x_n} < \sqrt{4} = 2$.

Increasing: $x_{n+1} > x_n \iff 2 + x_n > x_n^2 \iff (2 - x_n)(1 + x_n) > 0$, which holds because $0 < x_n < 2$.

Limit: by MCT, $x_n \rightarrow L$ with $1 \leq L \leq 2$. Square the recursion: $x_{n+1}^2 = 2 + x_n$. Take limits using the product and sum laws: $L^2 = 2 + L$, so $(L - 2)(L + 1) = 0$. Since $L \geq 1$, $L = 2$. $\square$

Full proof, step by step. Step 1 (compute and guess). $x_1 = 1$, $x_2 = \sqrt{3} \approx 1.73$, $x_3 = \sqrt{2 + \sqrt{3}} \approx 1.93$. The terms increase, apparently toward 2. (If $L = \sqrt{2 + L}$ then $L = 2$.)

Step 2 (bound by induction: $0 < x_n < 2$ for all n). Base: $0 < x_1 = 1 < 2$. Inductive step: assume $0 < x_n < 2$. Then $2 < 2 + x_n < 4$, and since $\sqrt{\cdot}$ is increasing, $0 < \sqrt{2} < \sqrt{2 + x_n} < \sqrt{4} = 2$, i.e. $0 < x_{n+1} < 2$.

Step 3 (increasing). Fix n . Both x_n and x_{n+1} are positive, so $x_{n+1} > x_n \iff x_{n+1}^2 > x_n^2$. Since $x_{n+1}^2 = 2 + x_n$:

$$x_{n+1} > x_n \iff 2 + x_n - x_n^2 > 0 \iff (2 - x_n)(1 + x_n) > 0.$$

By Step 2, $2 - x_n > 0$ and $1 + x_n > 0$, so the product is positive and $x_{n+1} > x_n$.

Step 4 (MCT). (x_n) is increasing and bounded above by 2, so by the Monotone Convergence Theorem it converges, say $x_n \rightarrow L$.

Step 5 (bounds on L). $1 = x_1 \leq x_n < 2$ for all n , so by the order property of limits $1 \leq L \leq 2$.

Step 6 (pass to the limit). The recursion gives $x_{n+1}^2 = 2 + x_n$ for all n . The shifted sequence (x_{n+1}) also converges to L , so by the product rule $x_{n+1}^2 \rightarrow L^2$. By the sum rule $2 + x_n \rightarrow 2 + L$. By uniqueness of limits, $L^2 = 2 + L$.

Step 7 (solve and discard). $L^2 - L - 2 = (L - 2)(L + 1) = 0$, so $L = 2$ or $L = -1$. Step 5 rules out -1 . Hence $x_n \rightarrow 2$. $\square$

Order matters: Step 6 is only valid *after* Step 4. Without knowing the limit exists, “ $L = \sqrt{2 + L}$ ” is meaningless.

Example 7.6 (A decreasing recursive sequence). $x_1 = 3$, $x_{n+1} = \sqrt{x_n + 2}$. *Bound:* $x_n > 2$ by induction ($x_{n+1} > \sqrt{4} = 2$). *Decreasing:* $x_{n+1} < x_n \iff x_n + 2 < x_n^2 \iff (x_n - 2)(x_n + 1) > 0$, which holds. By MCT, $x_n \rightarrow L \geq 2$ with $L^2 = L + 2$, so $L = 2$.

Full proof, step by step. Step 1 (compute and guess). $x_1 = 3$, $x_2 = \sqrt{5} \approx 2.24$, $x_3 = \sqrt{2 + \sqrt{5}} \approx 2.06$. The terms decrease, apparently toward 2.

Step 2 (bound by induction: $x_n > 2$ for all n). Base: $x_1 = 3 > 2$. Inductive step: assume $x_n > 2$. Then $x_n + 2 > 4$, so $x_{n+1} = \sqrt{x_n + 2} > \sqrt{4} = 2$.

Step 3 (decreasing). Both $x_n, x_{n+1} > 0$, so $x_{n+1} < x_n \iff x_{n+1}^2 < x_n^2 \iff x_n + 2 < x_n^2$, i.e.

$$x_n^2 - x_n - 2 = (x_n - 2)(x_n + 1) > 0.$$

By Step 2, $x_n - 2 > 0$ and $x_n + 1 > 0$, so this holds. Thus $x_{n+1} < x_n$.

Step 4 (MCT). (x_n) is decreasing and bounded below by 2. By the Monotone Convergence Theorem (decreasing version: it converges to $\inf\{x_n\}$), $x_n \rightarrow L$ for some L .

Step 5 (bounds on L). $2 < x_n \leq 3$ for all n , so by the order property $2 \leq L \leq 3$. (Strict $>$ becomes $\geq$ in the limit.)

Step 6 (pass to the limit). $x_{n+1}^2 = x_n + 2$. The left side tends to L^2 (shift and product rule). The right side tends to $L + 2$. So $L^2 = L + 2$.

Step 7 (solve and discard). $(L - 2)(L + 1) = 0$, so $L \in \{2, -1\}$, and Step 5 forces $L = 2$. $\square$

Compare with the previous example. Same recursion, but a different starting point. Starting above the fixed point 2 gives a decreasing sequence, and starting below it gives an increasing one. The bound you prove by induction always points toward the fixed point.

Example 7.7 (e exists). $a_n = (1 + \frac{1}{n})^n$ is increasing and bounded above by 3 (binomial expansion compared with $\sum \frac{1}{k!} \leq 1 + \sum \frac{1}{2^{k-1}}$). So it converges, and its limit is called e . **MCT proves existence without knowing the value.**

Full proof, step by step. Step 1 (expand with the binomial theorem).

$$a_n = \sum_{k=0}^n \binom{n}{k} \frac{1}{n^k}.$$

For $k \geq 1$,

$$\binom{n}{k} \frac{1}{n^k} = \frac{n(n-1)\cdots(n-k+1)}{k! n^k} = \frac{1}{k!} \cdot \frac{n}{n} \cdot \frac{n-1}{n} \cdots \frac{n-k+1}{n} = \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

So

$$a_n = \sum_{k=0}^n \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right),$$

where the empty product (for $k = 0$) is 1.

Step 2 (compare a_n with a_{n+1} term by term). Fix $k \leq n$. For each $j \geq 0$, $\frac{j}{n+1} \leq \frac{j}{n}$, so $1 - \frac{j}{n+1} \geq 1 - \frac{j}{n} \geq 0$ (as $j \leq k-1 < n$). All factors are nonnegative, so the products compare the same way:

$$\frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n+1}\right) \geq \frac{1}{k!} \prod_{j=0}^{k-1} \left(1 - \frac{j}{n}\right).$$

So term k of a_{n+1} is $\geq$ term k of a_n , for each $k = 0, \dots, n$.

Step 3 (increasing). a_{n+1} also has one extra term ($k = n+1$), which is ≥ 0 . Adding up Step 2 over $k = 0, \dots, n$ gives $a_{n+1} \geq a_n$.

Step 4 (drop the products). Each factor $1 - \frac{j}{n} \in [0, 1]$, so each product is ≤ 1 and $a_n \leq \sum_{k=0}^n \frac{1}{k!}$.

Step 5 ($k! \geq 2^{k-1}$ for $k \geq 1$). By induction: $1! = 1 = 2^0$, and if $k! \geq 2^{k-1}$ then $(k+1)! = (k+1)k! \geq 2 \cdot 2^{k-1} = 2^k$ (since $k+1 \geq 2$).

Step 6 (geometric bound).

$$a_n \leq 1 + \sum_{k=1}^n \frac{1}{k!} \leq 1 + \sum_{k=1}^n \frac{1}{2^{k-1}} = 1 + \frac{1 - (1/2)^n}{1 - 1/2} = 1 + 2\left(1 - \frac{1}{2^n}\right) < 3.$$

Step 7 (MCT). (a_n) is increasing (Step 3) and bounded above by 3 (Step 6). By the Monotone Convergence Theorem it converges. Its limit is *defined* to be e , and $2 = a_1 \leq e \leq 3$. $\square$

Example 7.8 (HW3 P4, averages). If (x_n) is increasing, then so is $y_n = \frac{x_1 + \dots + x_n}{n}$. The numerator of $y_{n+1} - y_n$ is $nx_{n+1} - S_n \geq 0$, because each $x_k \leq x_{n+1}$. (Positivity is not actually needed.)

Full proof, step by step. Let $S_n = x_1 + \dots + x_n$, so $y_n = \frac{S_n}{n}$ and $S_{n+1} = S_n + x_{n+1}$.

Step 1 (goal). Show $y_{n+1} - y_n \geq 0$ for every n .

Step 2 (common denominator).

$$y_{n+1} - y_n = \frac{S_{n+1}}{n+1} - \frac{S_n}{n} = \frac{nS_{n+1} - (n+1)S_n}{n(n+1)}.$$

Step 3 (simplify the numerator). Substitute $S_{n+1} = S_n + x_{n+1}$:

$$nS_{n+1} - (n+1)S_n = nS_n + nx_{n+1} - nS_n - S_n = nx_{n+1} - S_n.$$

Step 4 (rewrite as a sum of differences). nx_{n+1} is x_{n+1} added n times, and S_n has n terms, so

$$nx_{n+1} - S_n = \sum_{k=1}^n (x_{n+1} - x_k).$$

Step 5 (each difference is ≥ 0). Since (x_n) is increasing, $x_k \leq x_{k+1} \leq \dots \leq x_{n+1}$ for each $k \leq n$. (Formally, by chaining $x_k \leq x_{k+1}$ repeatedly, or by induction on $n+1-k$.) So every term $x_{n+1} - x_k \geq 0$, and the sum is ≥ 0 .

Step 6 (sign of the fraction). The numerator is ≥ 0 (Step 5) and the denominator $n(n+1) > 0$. So $y_{n+1} - y_n \geq 0$.

Step 7 (conclude). $y_n \leq y_{n+1}$ for all n , so (y_n) is increasing. $\square$

Intuition: adding a new value x_{n+1} that is at least as large as every earlier value cannot lower the average.

8 Preview: subsequences and Bolzano–Weierstrass (Week 5)

These may not be on Quiz 1, but you should know the statements.

- A **subsequence** is $(x_{n_k})_k$ with $n_1 < n_2 < \dots$.
- If $x_n \rightarrow L$ then every subsequence $\rightarrow L$. **So two subsequences with different limits imply divergence**, e.g. $(-1)^n$ with its even and odd terms.
- Every sequence has a monotone subsequence.
- **Bolzano–Weierstrass**: every bounded sequence has a convergent subsequence.
- **Cauchy**: $\forall \varepsilon \exists N: |x_n - x_m| < \varepsilon$ for all $n, m \geq N$. In $\mathbb{R}$, Cauchy $\iff$ convergent.

9 Common mistakes that cost points

1. **Writing ε – N proofs backwards.** Scratch work goes from the goal to N . The proof must go from “Let $\varepsilon > 0$ ” to “ $< \varepsilon$ ”.
2. **Choosing N that depends on n .** N may depend only on ε (and fixed constants).
3. **Forgetting to show that the sup is an upper bound.** Both parts (i) and (ii) are needed.
4. **Assuming the sup is in the set.** $\sup(0, 1) = 1 \notin (0, 1)$.
5. **Using limit laws when a limit doesn’t exist**, e.g. splitting $\frac{\sin n}{n}$ as $\lim \sin n \cdot \lim \frac{1}{n}$.
6. **Passing to the limit before proving convergence** in recursive sequences. MCT comes first, then solve $L = f(L)$.
7. **Multiplying an inequality by something that might be negative.**
8. **Induction without writing $P(n + 1)$ first**, and then “proving” the wrong target.
9. **Strict vs. non-strict**: limits preserve $\leq$ but not $<$.

10 Practice problems

Try these under quiz conditions. Solutions are in the next section.

Practice Problem 1. Prove by induction: $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ for all $n \in \mathbb{N}$.

Practice Problem 2. Prove by induction: $2^n \geq n + 1$ for all $n \in \mathbb{N}$.

Practice Problem 3. Prove that $\sqrt{3}$ is irrational.

Practice Problem 4. Prove $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Practice Problem 5. Find sup and inf of $S = \{(-1)^n (1 - \frac{1}{n}) : n \in \mathbb{N}\}$, say whether each is attained, and prove your claim about the sup.

Practice Problem 6. Find sup and inf of $T = \{\frac{n}{n+1} : n \in \mathbb{N}\}$ with proof.

Practice Problem 7. Let A be nonempty and bounded above, and $c > 0$. Prove $\sup(cA) = c \sup A$.

Practice Problem 8. Let $A \subseteq B$ be nonempty and bounded. Prove $\inf B \leq \inf A \leq \sup A \leq \sup B$.

Practice Problem 9. Use the definition to prove $\lim_{n \rightarrow \infty} \frac{2n^2 + 1}{n^2 + 3} = 2$.

Practice Problem 10. Use the definition to prove $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n}) = 0$.

Practice Problem 11. Prove $\lim_{n \rightarrow \infty} \frac{n^2}{n+1} = \infty$ from the definition.

Practice Problem 12. Prove that if $x_n \rightarrow L$ then $|x_n| \rightarrow |L|$. Is the converse true?

Practice Problem 13. Prove that $x_n = \frac{(-1)^n n}{n+1}$ diverges.

Practice Problem 14. Prove: if $x_n \rightarrow L$ and $x_n \geq 0$ for all n , then $L \geq 0$.

Practice Problem 15. Let $x_1 = 2$ and $x_{n+1} = \frac{x_n + 3}{2}$. Prove that (x_n) converges and find its limit.

Practice Problem 16. Compute (limit laws allowed): (a) $\lim \frac{5n^3 - n}{2n^3 + 7}$ (b) $\lim \frac{n+1}{n^2+1}$ (c) $\lim (\frac{2}{3})^n$
(d) $\lim \sum_{k=0}^n \frac{1}{2^k}$ (e) $\lim \frac{\cos(n^2)}{\sqrt{n}}$.

Practice Problem 17 (True or false: justify or give a counterexample).

- (a) If $x_n > 0$ for all n and $x_n \rightarrow L$, then $L > 0$.
- (b) Every bounded sequence converges.
- (c) Every convergent sequence is bounded.
- (d) If $x_n y_n \rightarrow 0$ then $x_n \rightarrow 0$ or $y_n \rightarrow 0$.
- (e) If (x_n) and (y_n) both diverge, then $(x_n + y_n)$ diverges.
- (f) Every nonempty subset of $\mathbb{Q}$ that is bounded above has a sup in $\mathbb{Q}$.
- (g) If $\sup S$ exists and $\sup S \in S$, then S has a maximum.

Practice Problem 18. Let A, B be nonempty and bounded above. Prove $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

Practice Problem 19. Find sup and inf of $\{\frac{1}{n} + \frac{1}{m} : n, m \in \mathbb{N}\}$.

Practice Problem 20. Prove that if $x_n \rightarrow 0$ and (y_n) is bounded, then $x_n y_n \rightarrow 0$. Use it to find $\lim \frac{(-1)^n n}{n^2 + 1}$.

Practice Problem 21. Prove that if $x_n \rightarrow \infty$ then $\frac{1}{x_n} \rightarrow 0$ (from the definitions).

Practice Problem 22. Use the definition to prove $\lim_{n \rightarrow \infty} \frac{3n + 1}{n - 2} = 3$ (consider $n \geq 3$).

Practice Problem 23. Prove that $\frac{n}{2^n} \rightarrow 0$.

Practice Problem 24. Solve $|2x - 1| < |x + 2|$.

Practice Problem 25. Prove: if $x_n \rightarrow L > 0$, then $x_n > \frac{L}{2}$ for all sufficiently large n .

Practice Problem 26. Show that $S = \{x \in \mathbb{R} : x^2 < 5\}$ is bounded above, explain why $\sup S$ exists without knowing its value, and identify $\sup S$.

11 Solutions

1. Base case: $1 = 1^2$. Target $P(n+1)$: $1 + 3 + \cdots + (2n-1) + (2n+1) = (n+1)^2$. By the hypothesis the left side is $n^2 + 2n + 1 = (n+1)^2$. $\square$

2. Base case: $2 \geq 2$. Assume $2^n \geq n+1$. Then $2^{n+1} = 2 \cdot 2^n \geq 2(n+1) = 2n+2 \geq n+2$. $\square$

3. Suppose $\sqrt{3} = p/q$ in lowest terms, so $p^2 = 3q^2$. If $3 \nmid p$, then $p = 3k \pm 1$ and $p^2 = 9k^2 \pm 6k + 1 = 3(3k^2 \pm 2k) + 1$ is not a multiple of 3, a contradiction. So $p = 3k$, which gives $9k^2 = 3q^2$, $q^2 = 3k^2$, and so $3 \mid q$. This contradicts lowest terms. $\square$

4. ($\subseteq$) Let $x \in A \cap (B \cup C)$. Then $x \in A$, and $x \in B$ or $x \in C$. Case $x \in B$: $x \in A \cap B$. Case $x \in C$: $x \in A \cap C$. Either way x is in the union. ($\supseteq$) If $x \in A \cap B$, then $x \in A$ and $x \in B \subseteq B \cup C$. The case $x \in A \cap C$ is the same. $\square$

5. The elements are $0, \frac{1}{2}, -\frac{2}{3}, \frac{3}{4}, -\frac{4}{5}, \dots$. Even n gives $1 - \frac{1}{n} \in [\frac{1}{2}, 1)$. Odd n gives $-(1 - \frac{1}{n}) \in (-1, 0]$. So $\sup S = 1$ and $\inf S = -1$, and *neither is attained*.

Proof that $\sup S = 1$: every element satisfies $(-1)^n(1 - \frac{1}{n}) \leq 1 - \frac{1}{n} < 1$, so 1 is an upper bound. Given $\varepsilon > 0$, choose an even n with $\frac{1}{n} < \varepsilon$ (Archimedean; if n is odd, use $n+1$). Then $1 - \frac{1}{n} \in S$ and $1 - \frac{1}{n} > 1 - \varepsilon$. $\square$ (The inf is symmetric, using odd n .)

6. $\frac{n}{n+1} = 1 - \frac{1}{n+1}$, which is increasing. So $\inf T = \min T = \frac{1}{2}$ (at $n = 1$), and $\sup T = 1$ is not attained. Upper bound: $\frac{n}{n+1} < 1$. Given $\varepsilon > 0$, pick n with $\frac{1}{n+1} < \varepsilon$. Then $\frac{n}{n+1} > 1 - \varepsilon$. $\square$

7. Let $\alpha = \sup A$. For $a \in A$: $a \leq \alpha$, so $ca \leq c\alpha$ (as $c > 0$), and $c\alpha$ is an upper bound of cA . Given $\varepsilon > 0$, $\frac{\varepsilon}{c} > 0$, so there is $a \in A$ with $a > \alpha - \frac{\varepsilon}{c}$. Then $ca \in cA$ and $ca > c\alpha - \varepsilon$. By Theorem 3.3, $\sup(cA) = c\alpha$. $\square$

8. Middle: pick any $a \in A$, so $\inf A \leq a \leq \sup A$. Right: $\sup B$ is an upper bound of $B \supseteq A$, so it is an upper bound of A . Hence $\sup A \leq \sup B$ ($\sup$ is the *least* upper bound). Left: the same argument with lower bounds. $\square$

9. $\left| \frac{2n^2+1}{n^2+3} - 2 \right| = \left| \frac{2n^2+1-2n^2-6}{n^2+3} \right| = \frac{5}{n^2+3} < \frac{5}{n^2} \leq \frac{5}{n}$. Let $\varepsilon > 0$ and choose $N > \frac{5}{\varepsilon}$. For $n \geq N$: $\left| \frac{2n^2+1}{n^2+3} - 2 \right| < \frac{5}{n} \leq \frac{5}{N} < \varepsilon$. $\square$

10. $\sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} < \frac{1}{\sqrt{n}}$. We want $\frac{1}{\sqrt{n}} < \varepsilon \iff n > \frac{1}{\varepsilon^2}$. Let $\varepsilon > 0$ and choose $N > \frac{1}{\varepsilon^2}$. For $n \geq N$: $|\sqrt{n+1} - \sqrt{n} - 0| < \frac{1}{\sqrt{n}} \leq \frac{1}{\sqrt{N}} < \varepsilon$. $\square$

11. For $n \geq 1$, $n+1 \leq 2n$, so $\frac{n^2}{n+1} \geq \frac{n^2}{2n} = \frac{n}{2}$. Given M , choose $N > 2M$. For $n \geq N$: $\frac{n^2}{n+1} \geq \frac{n}{2} \geq \frac{N}{2} > M$. $\square$

12. By the reverse triangle inequality, $||x_n| - |L|| \leq |x_n - L| < \varepsilon$ for $n \geq N$, with the same N that works for $x_n \rightarrow L$. The converse is *false*: $x_n = (-1)^n$ has $|x_n| \rightarrow 1$ but diverges. (It is true when $L = 0$, since $|x_n - 0| = ||x_n| - 0|$.) $\square$

13. Suppose $x_n \rightarrow L$. The even terms are $\frac{n}{n+1} \rightarrow 1$ and the odd terms are $-\frac{n}{n+1} \rightarrow -1$. Directly: take $\varepsilon = \frac{1}{2}$. For $n \geq N$ large, even n gives $x_n > \frac{1}{2}$ and odd n gives $x_n < -\frac{1}{2}$, so these terms are more than 1 apart. But $|x_n - x_m| \leq |x_n - L| + |L - x_m| < 1$, a contradiction. (Or: two subsequences with different limits.) $\square$

14. Suppose $L < 0$. Take $\varepsilon = -\frac{L}{2} > 0$. For large n , $x_n < L + \varepsilon = \frac{L}{2} < 0$, which contradicts $x_n \geq 0$. $\square$

15. The terms are $x_1 = 2$, $x_2 = 2.5$, $x_3 = 2.75$, so the guess is increasing with limit 3. *Bound* $x_n < 3$: true for $n = 1$; if $x_n < 3$ then $x_{n+1} < \frac{3+3}{2} = 3$. *Increasing*: $x_{n+1} - x_n = \frac{3-x_n}{2} > 0$ by the bound. By MCT, $x_n \rightarrow L$. Taking limits in $x_{n+1} = \frac{x_n+3}{2}$ gives $L = \frac{L+3}{2}$, so $L = 3$. $\square$

16. (a) $\frac{5}{2}$ (b) 0 (lower degree on top) (c) 0 since $|\frac{2}{3}| < 1$ (d) $\frac{1}{1-1/2} = 2$ (e) 0, since $\left| \frac{\cos(n^2)}{\sqrt{n}} \right| \leq \frac{1}{\sqrt{n}} \rightarrow 0$ (squeeze).

17.

- (a) False: $\frac{1}{n} > 0$ but $\frac{1}{n} \rightarrow 0$. (You only get $L \geq 0$.)
- (b) False: $(-1)^n$. (It *does* have a convergent subsequence, by B-W.)
- (c) True: the “convergent $\Rightarrow$ bounded” theorem.
- (d) False: $x_n = 1, 0, 1, 0, \dots$ and $y_n = 0, 1, 0, 1, \dots$ have $x_n y_n = 0$, but neither converges to 0.
- (e) False: $x_n = (-1)^n$ and $y_n = -(-1)^n$ sum to 0.
- (f) False: $\{q \in \mathbb{Q} : q^2 < 2\}$ has no sup in $\mathbb{Q}$, since the sup would be $\sqrt{2}$. This is exactly why $\mathbb{R}$ needs the completeness axiom.
- (g) True: $\sup S$ is an upper bound in S , which is the definition of a maximum.

18. Let $\alpha = \sup A$, $\beta = \sup B$, $M = \max\{\alpha, \beta\}$. *Upper bound*: if $x \in A \cup B$, then $x \leq \alpha \leq M$ or $x \leq \beta \leq M$. *Least*: say $M = \alpha$. Given $\varepsilon > 0$, there is $a \in A \subseteq A \cup B$ with $a > \alpha - \varepsilon = M - \varepsilon$. $\square$

19. $\sup = \max = 2$ (at $n = m = 1$), since $\frac{1}{n} + \frac{1}{m} \leq 1 + 1$. $\inf = 0$, not attained: 0 is a lower bound since every element is > 0 . Given $\varepsilon > 0$, pick n with $\frac{1}{n} < \frac{\varepsilon}{2}$. Then $\frac{1}{n} + \frac{1}{n} < \varepsilon$, with $m = n$. $\square$

20. Let $|y_n| \leq M$ with $M > 0$. Given $\varepsilon > 0$, choose N with $|x_n| < \frac{\varepsilon}{M}$ for $n \geq N$. Then $|x_n y_n| \leq M|x_n| < \varepsilon$. Application: $\frac{(-1)^n n}{n^2+1} = (-1)^n \cdot \frac{n}{n^2+1}$, where $(-1)^n$ is bounded and $0 < \frac{n}{n^2+1} < \frac{1}{n} \rightarrow 0$. So the limit is 0. $\square$

21. Let $\varepsilon > 0$. Apply $x_n \rightarrow \infty$ with $M = \frac{1}{\varepsilon}$: there is N with $x_n > \frac{1}{\varepsilon} > 0$ for $n \geq N$. Then $0 < \frac{1}{x_n} < \varepsilon$, so $\left| \frac{1}{x_n} - 0 \right| < \varepsilon$. $\square$

22. $\frac{3n+1}{n-2} - 3 = \frac{7}{n-2} > 0$ for $n \geq 3$. For $n \geq 3$ we have $n-2 \geq \frac{n}{3}$ (equivalent to $\frac{2n}{3} \geq 2$), so $\frac{7}{n-2} \leq \frac{21}{n}$. Given $\varepsilon > 0$, choose $N > \max\{3, \frac{21}{\varepsilon}\}$. For $n \geq N$: $\left| \frac{3n+1}{n-2} - 3 \right| \leq \frac{21}{n} \leq \frac{21}{N} < \varepsilon$. $\square$

23. By the binomial theorem, $2^n \geq \binom{n}{2} = \frac{n(n-1)}{2}$ for $n \geq 2$, so $0 < \frac{n}{2^n} \leq \frac{2}{n-1}$. Given $\varepsilon > 0$, choose $N > 1 + \frac{2}{\varepsilon}$. For $n \geq N$: $\frac{n}{2^n} \leq \frac{2}{n-1} < \varepsilon$. $\square$

24. Both sides are ≥ 0 , so the inequality is equivalent to comparing squares: $(2x - 1)^2 < (x + 2)^2 \iff 3x^2 - 8x - 3 < 0 \iff (3x + 1)(x - 3) < 0 \iff -\frac{1}{3} < x < 3$. (Alternatively, split into cases at $x = \frac{1}{2}$ and $x = -2$.)

25. Take $\varepsilon = \frac{L}{2} > 0$. There is N with $|x_n - L| < \frac{L}{2}$ for $n \geq N$, so $x_n > L - \frac{L}{2} = \frac{L}{2}$. **This is the “eventually positive” lemma, used in the quotient rule.** $\square$

26. If $x \geq 3$ then $x^2 \geq 9 > 5$, so every $x \in S$ satisfies $x < 3$, and 3 is an upper bound. $S \neq \emptyset$ since $0 \in S$. By the Completeness Axiom, $\sup S$ exists, even though we haven’t computed it. In fact $\sup S = \sqrt{5}$, whose existence was proved in lecture by exactly this sup argument for $\sqrt{2}$.